

Reproducibility Assessment - Kiwi Group

Anna Rosa Klaver Are Markku Alanen Tjihkkom
Jemima Agnew Liradie Allen Naomi Takizawa

2026-02-24

Reproducibility Assessment

This report assesses the reproducibility of the findings from *Comparing the cognitive load of gesture and action production: a dual-task study* by Hostetter and Bahl (2023).

Summary

In this article, the authors investigated whether speech-accompanying gestures and speech-accompanying actions (which refer to moving wooden pieces on a grid in this study) have similar or different effects on a speaker's cognitive load, compared to speaking with hands still.

We focus on the first of two experiments described in the article. In Experiment 1, 49 participants completed 24 trials in a within-subject dual-task paradigm. On each trial, they first saw a 3×3 grid containing letters placed in specific locations, along with a spatial pattern that remained visible on the screen. The primary task was to describe the pattern, and the secondary task was to remember both the letters (measuring verbal load) and their locations in the grid (measuring spatial load). The 24 trials were run under three conditions (gesture, action and hand-still), each consisting of 8 trials. After each description of the pattern (primary task), participants reported letters and their locations (secondary task). Then, the authors tested whether different conditions predicted memory performance while controlling for speech characteristics seen during the descriptions (word count, duration, filled pauses, and deictic references). The same procedure was done in experiment 2 with the difference being having the spatial pattern disappear after 5 seconds, which forced the participants to describe the pattern from memory. Due to time constraints and the number of models in the full paper, our computational and analytical report focus only on Experiment 1.

The authors make two main claims for Experiment 1: one about Gesture vs. Hand-still and another about Action vs. Gesture.

The first claim is: “compared with keeping hands still, gestures reduced verbal load when the pattern remained visible on the screen.” The authors provided the following statistics generated by linear mixed models to support the claim: gesture: $M = 2.96$, $SE = 0.14$; hands-still: $M = 2.73$, $SE = 0.14$; $\Delta = .24$, $SE = .09$, $t = 2.53$, $p = 0.01$. This indicates speech-accompanying gestures reduces verbal cognitive load compared to when keeping hands still, and this difference in cognitive load is statistically significant. They did not find significant differences between gesture, action, and hands-still for spatial tasks (recalling locations), concluding that the effect was specific for verbal load.

The second claim that the authors drew from Experiment 1 (and also from Experiment 2) is that “there was no difference in the cognitive load associated with gesture and action.” This is supported by the following results: regarding verbal memory, performance with speech-accompanying action ($M = 2.88$, $SE = 0.14$) does not significantly differ from that with keeping hands still ($\Delta = .15$, $SE = .10$, $t = 1.54$, $p = 0.12$), or with speech-accompanying gesture ($\Delta = .08$, $SE = .10$, $t = 0.82$, $p = 0.41$). This trend was also seen in Experiment 2, and the authors concluded that there is not a significant difference between action and gesture in the context of this study.

Finally, in the conclusion section, the authors made an overarching claim that “moving the hands in meaningful ways can benefit verbal load.” This claim seems to be based on the following findings: (1) gesture reduced verbal load compared to hands-still condition in Experiment 1, (2) action reduced verbal load compared to hands-still condition in Experiment 2, (3) no significant difference found between gesture and action conditions in neither experiments, (4) no significant effect of any of the conditions on spatial load was found in neither experiments. Moving hands can relieve verbal cognitive load, but whether gestures or actions are the more effective way for doing so seems to depend on the context.

Analytical Assessment

The analytical and computational assessment focuses on reproducing Experiment 1 of Hostetter and Bahl (2023). This included a reproduction of data cleaning and exclusion, descriptive statistics and three model sets, including an AIC-based model comparison. We were unable to reproduce the demographic summaries as participant demographic data was not included in the osf repository.

Data cleaning and descriptives.

Hostetter and Bahl (2023) report an analysed sample of 49 participants after excluding 4 participants due to technical issues and 7 participants who were not native english speakers. This matches the participants ($n=49$) included in the available dataset. Hostetter and Bahl (2023) provide a detailed list of reasons for trial exclusions by condition and these are coded into the data set as a single inclusion factor. They filtered the data to include only the trials where the participants correctly followed the instructions. In the paper, this is said to exclude 10.6% of trials (i.e., 125 trials). In our reproduction, this resulted in the exclusion of 10.8% of

trials (127 trials). The authors report the mean trials and range per participant per condition. Our reproduction closely matches the reported means for two of the conditions: Gesture and Hands still. However, for the Make condition, our overall mean ($m=6.37$) is lower than the reported mean ($m=6.50$). The paper reports a range of 3-8 trials. In our filtered dataset, Participant 2 has 0 Make trials after exclusions. This is likely the source of the discrepancy. Despite having no valid Make condition trials, Participant 2's trials for the other conditions do not appear to be excluded from the authors' data analysis.

Model Set 1: Effects of condition on speech measures.

In Model Set 1, the analysis examined whether the trial condition (gesture, make, or hands still) affected different speech measures. The four speech measures included word count, filled pauses, time spent speaking, and deictic reference. Hostetter and Bahl (2023) initially included random slopes for Condition by Participant and Item. The authors report dropping slopes if a model did not converge. In our reproduction, the full model for word count by trial condition converged. The full models for time spent speaking and filled pauses provided singular fit issues. The model for deictic reference provided non-convergence errors. After simplifying the random slopes for these three model sets, we were able to make all three models converge. The final random-effects structure for the models was not reported in [cite]. As such, there remains some ambiguity as to whether our model structure exactly matched those used in the article's analysis. To test the significance of pairwise comparisons between levels of condition, the authors used the `diffsmeans` function from the `lmerTest` package (Kuznetsova et al., 2017; Hostetter & Bahl, 2023). We were unfamiliar with this function and did not reproduce these significance tests.

Model Set 2: Effects of speech measures on memory

In Model Set 2, the analysis examined whether each speech measure had an effect on the Letter and Grid location memory measure. When describing the model structure in the paper, they state that they "entered the measure as a fixed factor" (Hostetter & Bahl, 2023). We found this wording ambiguous. It was unclear whether they had (a) 8 separate models (with each speech variable separately), or (b) 2 combined models (with all four speech predictors simultaneously). As such, we ran models as interpreted in both options (a) and (b). For this set of models, the authors did not state whether they used random slopes or just intercepts, nor did they mention any convergence issues. We started with a maximal random effects structure. For the option (a) models, we had to simplify the random effects structure of most models due to singular fit issues. The same was required for the two option (b) models, one due to singular fit issues and the other due to non-convergence. In the article, they then 'determined significance with the `lmerTest` package' (Hostetter & Bahl, 2023). Again, we found this statement to be a bit vague as to how they actually decided to methodologically 'determine significance'.

Model Set 3: Effects of Condition (Gesture, Hands Still, and Make) on cognitive load

In Model Set 3, two AIC-based model comparisons were used to identify the best-fitting model predicting the effect of task condition on memory performance (either letters recalled

or locations recalled). For both letters and locations recalled, the authors fit a full model including trial condition and four speech measure predictors. Fourteen nested models with different combinations of speech measure predictors plus Condition were made, and the best-fitting model was selected. The AIC model selection was reproducible and the best-fit models we found matched those reported in the article. The pairwise differences we produced matched those reported in the paper. Further significance testing was not mentioned in the article’s description. The authors have presented the analysis as confirmatory, but the AIC-based model comparisons lean towards an exploratory analysis. The analysis method they used is functionally exploratory as they look for patterns or findings without a pre-existing theoretical justification. However, the wording they use to describe their analysis posits it as confirmatory. They use the data to predict a hypothesis that gives findings based on said data, even though they should use the data to predict the probability of their hypothesis.

From our analytical assessment, we were able to reproduce the overall core findings from Experiment 1 of Hostetter and Bahl (2023). We were unable to complete an exact replication based on the analysis description alone due to some small discrepancies between our data and the reported data, some ambiguous methodological description, and under-specified random effects. The data-driven model selection in Model Set 3 leans more towards an exploratory analysis, rather than hypothesis-driven.

Computational Assessment

The code provided by the authors does not run on a clean machine. Firstly, libraries are loaded without instructions to install them or specifications about what they are. In order to run the code from a clean machine, a command to install missing libraries should be provided. Secondly, the data is not loaded in, as the code does not specify which file to read. We therefore modified the line to load in data from `data<-AGI_study_1_data` to `data<-read.csv("AGI_study_1_data.csv")` to read the provided .csv file. After loading in libraries and loading the data correctly, the rest of the code runs.

The code runs mostly in line with the author’s comments, save a few discrepancies. In line 52, the output message gives a warning about the model not converging, which is not the same as a singular fit, although the comment on line 56 states so. When making the violin graph for average letters recalled, published in the article as Figure 2, the output gives the following warning message: *Removed 48 rows containing missing values or values outside the scale range ('geom_violin()')*. The authors do not explain this, and it took us some time to interpret that the error message is simply due to the graph limit specification `ylim(0,5)`. The “removed 48 rows” refers to the top of the violin graph being cut off at $y=5$, which can be seen in the “Make” condition in Figure 2 in the article, and seems harmless. However, for the reproducer’s comfort, this should have been explained. It would also have been useful to mention that the printed violin graph differs in each iteration due to how R creates it. The plot points get a new position on the x-axis every time the plot is printed and may thus seem different from the plot in Figure 2, however, this does not matter, as their position on the y-axis remains

consistent, which is the important part. Optimally, this should be mentioned as not to confuse the reproducers.

The script in general is hard to understand for outsiders. For example, what is consistently referred to as “letters” in the script is used to predict the effect of gesture on retained verbal information, while “locations” predicts the effect on retained spatial information. It would be easier if the labels are specified, as inconsistent wording requires more time to assess how well the script and the results in the article match. We see the issue of unclarity also in the report of the numbers in Table 1 in the article, the means and standard errors of the speech variables in the experiment conditions, as these are not directly identifiable in the script output. In other words, we have not found where the information in Table 1 comes from. The identification of the script output in the article is further complicated by both unclear labels and column names. For example, our interpretation is that the data for the article variable “Deictic speech” are calculated from the data in the column “index.speech”. The condition names in the data, “M”, “G” and “NG” are for a long time not elaborated, and only in printing the plot are they changed to “Make”, “Gesture” and “Hands Still”, which are the conditions referenced in the article. This discrepancy makes it harder to identify which the conditions the article references when trying to recognise them in the script.

Two more participants are excluded than the authors report. In section 2.1.4, the authors state that 125 trials were excluded, leading to the loss of 10.6% of the data. However, the script excludes 127 trials, leading to the loss of 10.8% of the data. The two additional exclusions seem to be due to them being marked as excluded in the data set, and except for this one instance of discrepancy, the report from the authors is based on excluding 127 trials. We assume that this is caused by a human error in presenting the information in section 2.1.4, as the rest of the reported findings have no discrepancies.

Finally, the script prints two graphs, Figure 2, as already mentioned, as well as a violin plot for average locations recalled, which was reported to have no significant results. It is commendable that the authors keep exploratory code in the script, as this helps us better trace what they did and get a fuller overview of their decisions.

Final Evaluation

We found the claims in Experiment 1 of Hostetter and Bahl (2023) to be broadly reproducible and trustworthy. We were able to reproduce the main patterns of results using the available dataset and scripts. When modelling choices were not adequately described in the paper, we were able to reconstruct similar results using reasonable modelling decisions.

However, the analytical description was not fully transparent. Small discrepancies in the reported exclusions, under-specification of random-effects structures, and ambiguous model descriptions reduce clarity in reproduction. The data-driven model selection was described in a manner that alluded to conventional hypothesis testing. These issues somewhat weaken the strength of the claims of Hostetter and Bahl (2023); however, they do not necessarily

invalidate the findings. The conclusions of the experiment are plausible and moderately reproducible. Greater transparency regarding model specification is needed, as well as an explicit acknowledgment of the exploratory nature of the analysis. This would greatly strengthen the credibility of Hostetter and Bahl's (2023) claims.

References

Hostetter, A. B., & Bahl, S. (2023). Comparing the cognitive load of gesture and action production: A dual-task study. *Language and Cognition*, 15(3), 601–621.

Kuznetsova, A., Brockhoff, P. B., & Christensen, R. H. B. (2017). *lmerTest package: Tests in linear mixed effects models*. *Journal of Statistical Software*, 82(13), 1–26.